

MINSI LU

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EDUCATION BACKGROUND

Department of Electrical & Computer Engineering, University of Waterloo	Waterloo, Canada
Doctor of Philosophy	<i>Sep. 2025 – Present</i>
Department of Pharmacy science, Tsinghua University	Beijing, China
Bachelor of Science	<i>Aug. 2019 - Jul. 2024</i>
Paul G. Allen School of Computer Science and Engineering, University of Washington	Seattle, US
Visiting Student	<i>Jun. 2023- Oct. 2023</i>

PUBLICATIONS

Bowen Gao, Bo Qiang, Haichuan Tan, Minsi Ren, Yinjun Jia, **Minsi Lu**, Jingjing Liu, Weiying Ma, Yanyan Lan, ‘**DrugCLIP: Contrastive Protein-Molecule Representation Learning for Virtual Screening**’, **NeurIPS 2023**.

Tangqi Fang, Yifeng Liu, Addie Woicik, **Minsi Lu**, Anupama Jha, Xiao Wang, Gang Li, Borislav Hristov, Zixuan Liu, Hanwen Xu, William S. Noble, Sheng Wang, ‘**Enhancing Hi-C Contact Matrices for Loop Detection with Capricorn**’, **ISMB 2024**.

RESEARCH EXPERIENCE

Research Intern, ERA Environmental Software Solution	Montreal, Canada
Project: LLM-Based Software Testing and Web Agents	<i>Sep. 2026 – Present</i>
Worked on LLM-based software engineering, focusing on Playwright/Selenium test generation and repair, LLM-based web agents for browser interaction, and automated testing workflows in CI pipelines.	
Research Assistant, School of Computer Science & Engineering, University of Washington	Seattle, US
Project: Resolution Enhancement and Time Dimension Modeling for Hi-C Data	<i>Jun. 2023 - Oct. 2023</i>
<i>Advisor: Prof. William Stafford Noble, Prof. Sheng Wang</i>	
- Developed Capricorn, a diffusion-based framework for Hi-C resolution enhancement that integrates high-order chromatin features as additional views of the contact matrix to generate high-resolution Hi-C data; the work was published at ISMB 2024. - The link is https://academic.oup.com/bioinformatics/article/40/Supplement_1/i471/7700867	
Research Assistant, Institute for AI Industry Research	Beijing, China
Project: Contrastive Learning for Drug Virtual Screening	<i>Feb. 2023 - Jun. 2023</i>
<i>Advisor: Prof. YanYan Lan</i>	
- Developed DrugCLIP, a contrastive learning framework that reformulates virtual screening as a dense retrieval task and aligns protein-pocket and molecular representations using large-scale pairwise data without explicit binding-affinity labels; the work was published at NeurIPS 2023. - The link is https://neurips.cc/virtual/2023/poster/70609	

TEACHING EXPERIENCE

Teaching Assistant University of Waterloo	
MTE 241: Computer Structures and Real-Time Systems.	Fall 2025
Led tutorial sessions, held office hours, and graded coursework.	

SKILLS & LANGUAGES

Mathematics: Calculus, Linear Algebra, Probability and Statistics.

Programming Skills: Skilled at C/C++, Java, Python, R, SQL. Familiar with Linux/Unix

Machine Learning & AI: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn; Experienced in Deep Learning architecture design and deployment.

Systems & Infrastructure: Linux/Unix, Git, Docker, Kubernetes; Cloud Computing(AWS) and CI/CD.

Languages: Mandarin (native), English (fluent), French

AWARDS & HONORS

Scholarship for Excellent in Academic, University Scholarship (10%)	2023
First Prize in 16 th "Novozymes Cup" Life Science Culture Festival, Municipal Award (10%)	2021
Third Prize in 39 th "Challenges Cup" Student Academic Competition, University Award	2021
Scholarship for Progress in Academic, University Scholarship	2021
Third Prize in Winter Vacation Social Practice for Students, University Award	2021
Gold Medal in 27 th China Biology Olympiad(CBO), National Award	2018